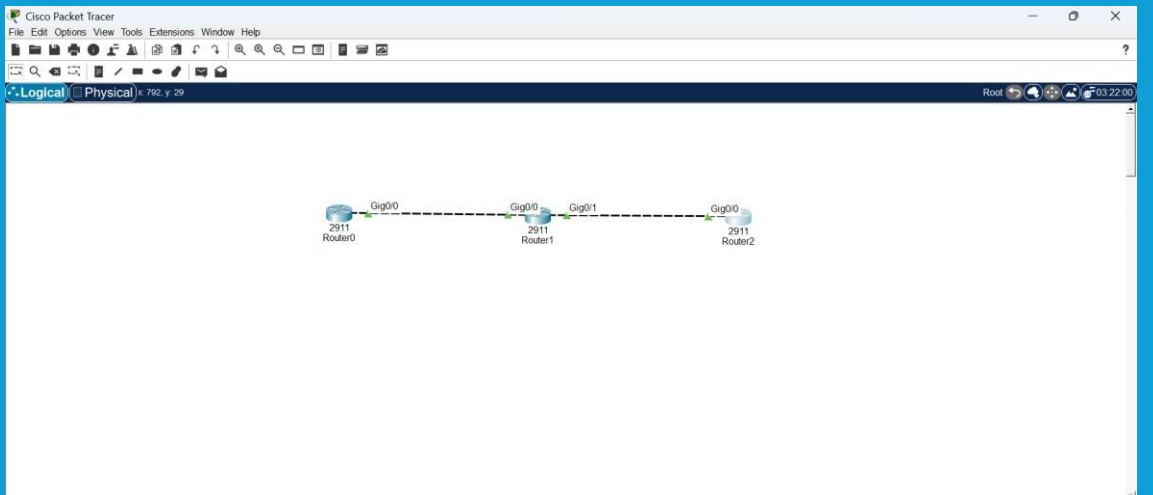


1.Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.



The screenshot shows a network topology in Cisco Packet Tracer. Three routers, labeled 2911, are connected in a line. Router0 is connected to Router1 via GigabitEthernet0/0 and GigabitEthernet0/1. Router1 is connected to Router2 via GigabitEthernet0/0 and GigabitEthernet0/1. The routers are labeled 2911. The interface names are Gig0/0 and Gig0/1.

```
Router0>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.0.1 YES manual up up
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 unassigned YES unset administratively down down
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

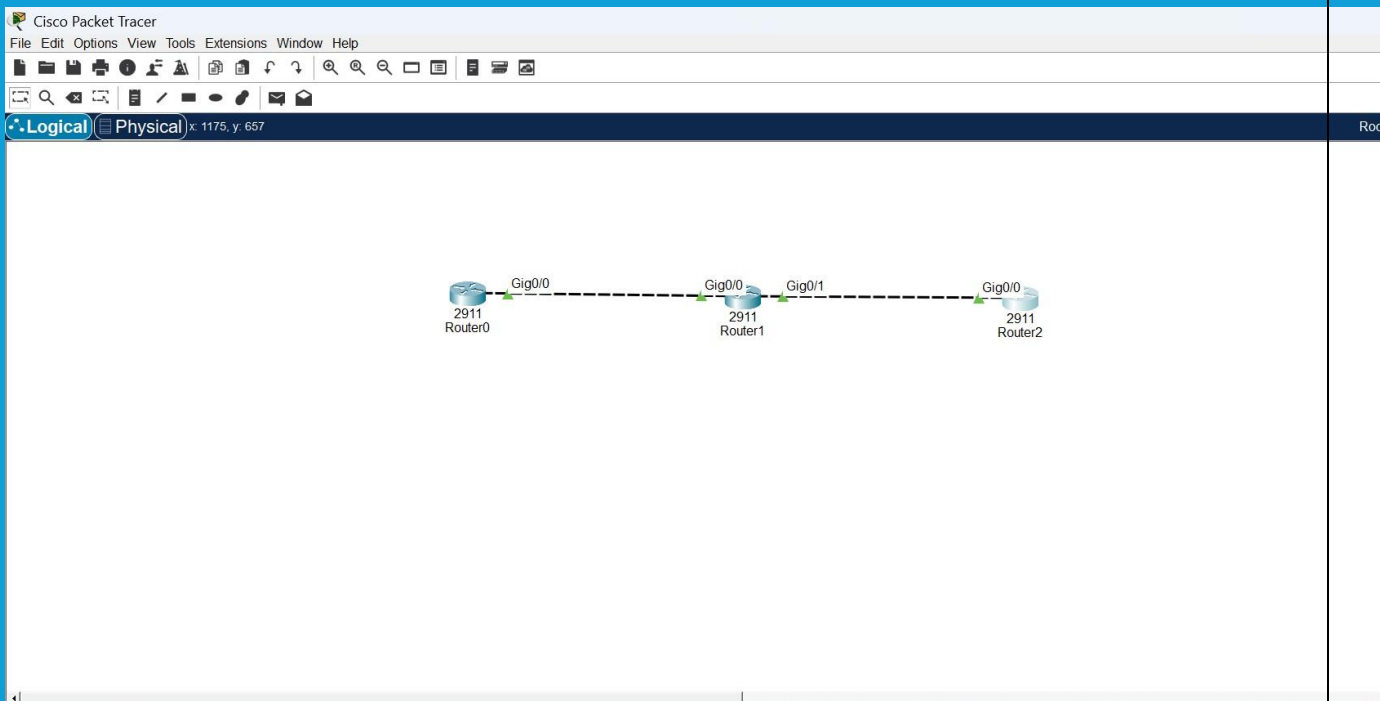
```
Router1>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 20.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.0.2 YES manual up up
GigabitEthernet0/1 20.0.0.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 unassigned YES unset administratively down down
Serial0/0/1 unassigned YES unset administratively down down
Serial0/1/0 unassigned YES unset administratively down down
Serial0/1/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 20.0.0.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 20.0.0.2 YES manual up up
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 unassigned YES unset administratively down down
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```

2.Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.



```
Router0

Physical Config CLI Attributes

IOS Command Line Interface

GigabitEthernet0/1      unassigned      YES unset      administratively down down
GigabitEthernet0/2      unassigned      YES unset      administratively down down
Serial0/0/0             unassigned      YES unset      administratively down down
Serial0/0/1             unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
R       20.0.0.0/24 is subnetted, 1 subnets
        20.0.0.0/24 [120/1] via 10.0.0.2, 00:00:00, GigabitEthernet0/0
Router#
```

```
Router1

Physical Config CLI Attributes

IOS Command Line Interface

Serial0/1/0             unassigned      YES unset      administratively down down
Serial0/1/1             unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#network 20.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
L       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
Router#
```

```
Router2

Physical Config CLI Attributes

IOS Command Line Interface

Serial0/0/0             unassigned      YES unset      administratively down down
Serial0/0/1             unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 20.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
R       10.0.0.0/24 [120/1] via 20.0.0.1, 00:00:12, GigabitEthernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/0
L       20.0.0.2/32 is directly connected, GigabitEthernet0/0
Router#
```

3. Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

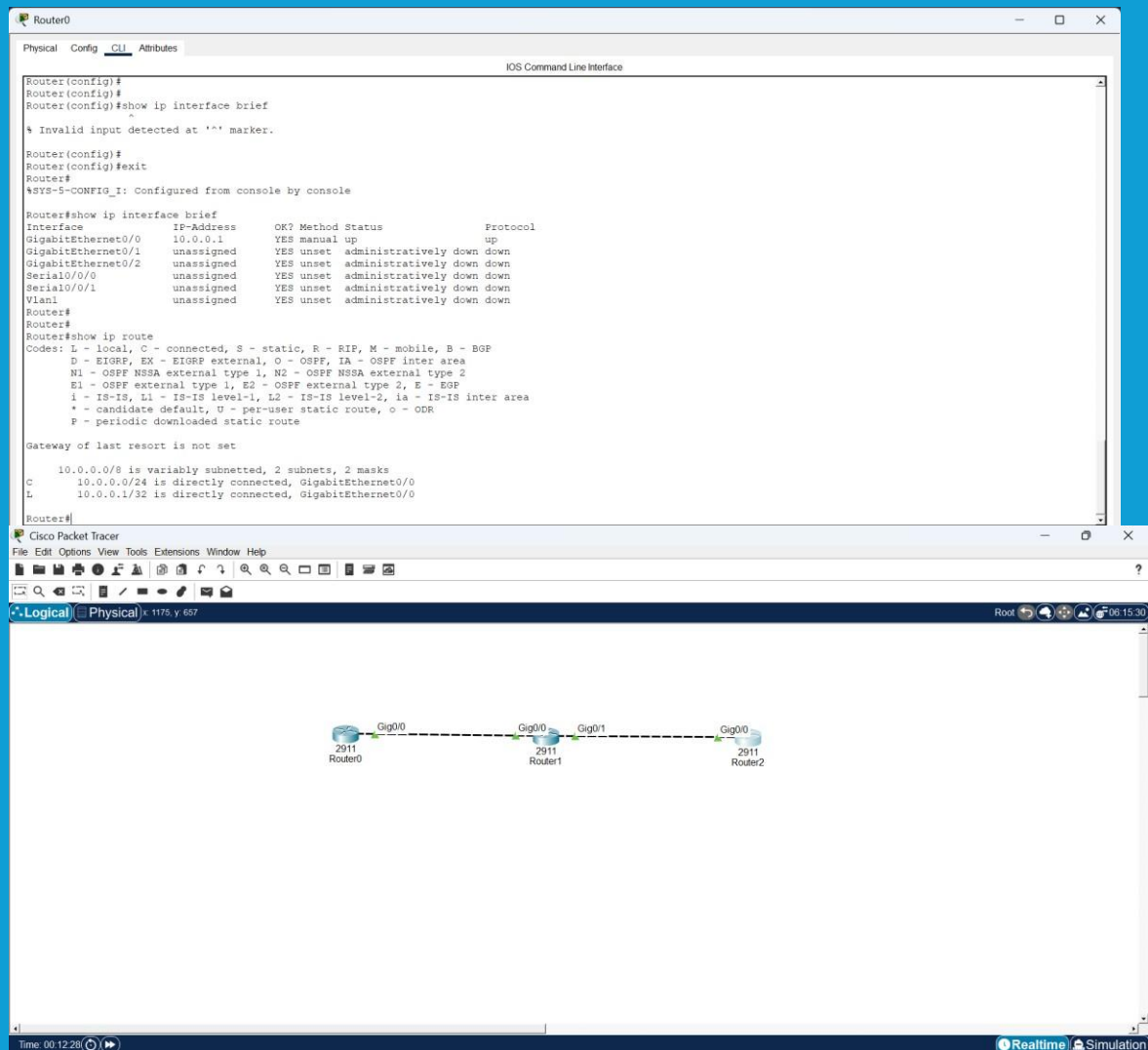


The top window shows a network topology with four routers: Router0, Router1, Router3, and Router2. Router0 is connected to Router1 via GigabitEthernet0/0 and GigabitEthernet0/1. Router1 is connected to Router3 via GigabitEthernet0/0 and GigabitEthernet0/1. Router3 is connected to Router2 via GigabitEthernet0/0. The bottom window shows the configuration of Router3.

```
Router3
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#ip address 30.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
*LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#interface g0/0
Router(config-if)#ip address 20.0.0.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface g0/1
Router(config-if)#ip address 30.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 20.0.0.2 YES manual up up
GigabitEthernet0/1 30.0.0.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 unassigned YES unset administratively down down
Serial0/0/1 unassigned YES unset administratively down down
Serial0/1/0 unassigned YES unset administratively down down
Serial0/1/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
```



4. Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between

routers.

Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.



The screenshot displays the Cisco Packet Tracer interface. The top section shows a network topology with four routers: Router0, Router1, Router3, and Router2. Router0 is connected to Router1 via GigabitEthernet0/0 and GigabitEthernet0/1. Router1 is connected to Router3 via GigabitEthernet0/1 and GigabitEthernet0/0. Router3 is connected to Router2 via GigabitEthernet0/0 and GigabitEthernet0/1. The bottom section shows the configuration of Router0 in the CLI. The configuration includes enabling RIP version 2, disabling auto-summary, and advertising the 10.0.0.0 network.

```
Router>
Router>
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#end
Router#
Router#
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip route
```

